

Descriptive Statistics Primer

A Study of the Iris Dataset using Python & Pandas

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Task: Task 9 – Descriptive Statistics Primer
Dataset: Iris Dataset (150 records, 4 numeric columns)
Tools Used: Python, Pandas (Google Colab)

1. Description

This task involved calculating and interpreting the mean, median, mode, standard deviation, and percentiles for the key numeric columns of the Iris dataset, in order to understand what each statistic reveals about the shape and spread of the data.

2. Objective

To build intuition for when to use each descriptive statistic and what it reveals about a distribution — specifically, how comparing these values can uncover hidden structure (such as sub-groups) within a dataset.

3. Summary Statistics Table

The table below shows the summary statistics computed for four numeric columns of the Iris dataset (sepal_length, sepal_width, petal_length, petal_width) using pandas' describe() and mode() functions.

Statistic	sepal_length	sepal_width	petal_length	petal_width
Count	150	150	150	150
Mean	5.84	3.06	3.76	1.20
Median (50%)	5.80	3.00	4.35	1.30
Mode	5.0	3.0	1.4	0.2
Std. Deviation	0.83	0.44	1.77	0.76
25th Percentile	5.10	2.80	1.60	0.30
75th Percentile	6.40	3.30	5.10	1.80
Min	4.30	2.00	1.00	0.10
Max	7.90	4.40	6.90	2.50

4. Species-wise Mean Comparison

To investigate the difference observed between mean, median and mode for the petal measurements, the data was grouped by the 'species' column. The resulting group-wise means are shown below.

Species	sepal_length	sepal_width	petal_length	petal_width
Setosa	5.006	3.428	1.462	0.246
Versicolor	5.936	2.770	4.260	1.326
Virginica	6.588	2.974	5.552	2.026

5. Write-up: What the Numbers Reveal

I calculated descriptive statistics for the 4 numeric columns in the Iris dataset — sepal_length, sepal_width, petal_length, and petal_width.

- For sepal_length and sepal_width, the mean, median, and mode values are all quite close to each other. This shows that their distribution is roughly symmetric, without any major skew in these two columns. The standard deviation is also low (0.83 and 0.44), meaning the values stay fairly close to their average.
- For petal_length and petal_width, I noticed a bigger difference between the mean, median, and mode. For example, petal_length had a mean of 3.76 but a mode of only 1.4 — at first this looked like the data was skewed. But when I grouped the data by species using `groupby('species')`, it became clear that there are actually 3 different species with very different petal sizes:
 - Setosa has the smallest petals (length ~1.46, width ~0.25)
 - Versicolor is medium-sized (length ~4.26, width ~1.33)
 - Virginica has the largest petals (length ~5.55, width ~2.03)
- Looking at the 25th and 75th percentiles also confirms this — the middle 50% of petal_length values (1.6 to 5.1) is a much wider range compared to sepal_width (2.8 to 3.3), which shows petal measurements are far more spread out across the different species.
- So the gap wasn't due to skewness in a single distribution — it was because data from three different groups was mixed together.

My Takeaway: *If mean, median, and mode differ a lot for a column, it's not always correct to assume the data is skewed. It's important to first check whether there's a hidden category or group in the data causing that pattern. In the Iris dataset, species turned out to be exactly that hidden factor.*